

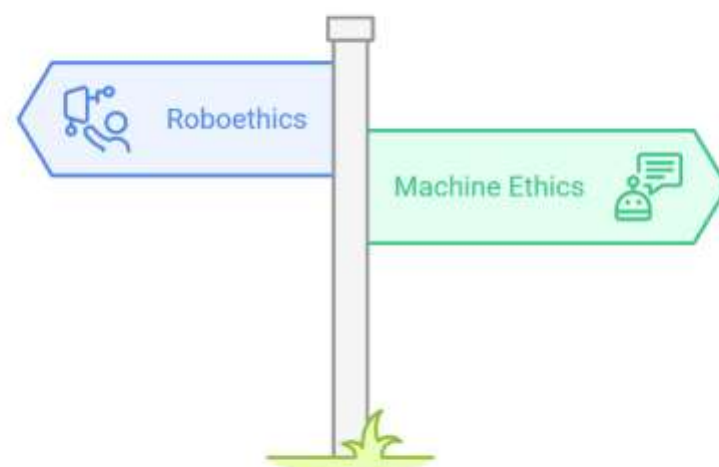
INTRODUCTION T O AI AND APPL ICATIONS

Module 4

Module 4: Current Trends in Artificial Intelligence

8.1 AI and Ethical Concerns

Understanding AI Ethics and Its Importance AI ethics addresses the moral obligations and responsibilities of AI systems, their creators, and their users. It involves ensuring that AI technologies align with human values and societal



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Robo Ethics vs. Machine Ethics

Aspect	Robo Ethics	Machine Ethics
Focus	Human behaviour regarding robots	Robot's own decision-making
Who follows the ethics?	Humans (designers, users, companies)	Machines/AI systems
Main concern	Safe creation & use of robots	Embedding morality into algorithms
Type of questions	"Should we build this robot?"	"How should the robot decide in a dilemma?"
Time of application	Before the robot is deployed	During the robot's operation
Example	Laws for robot use	Ethical rules inside self-driving cars

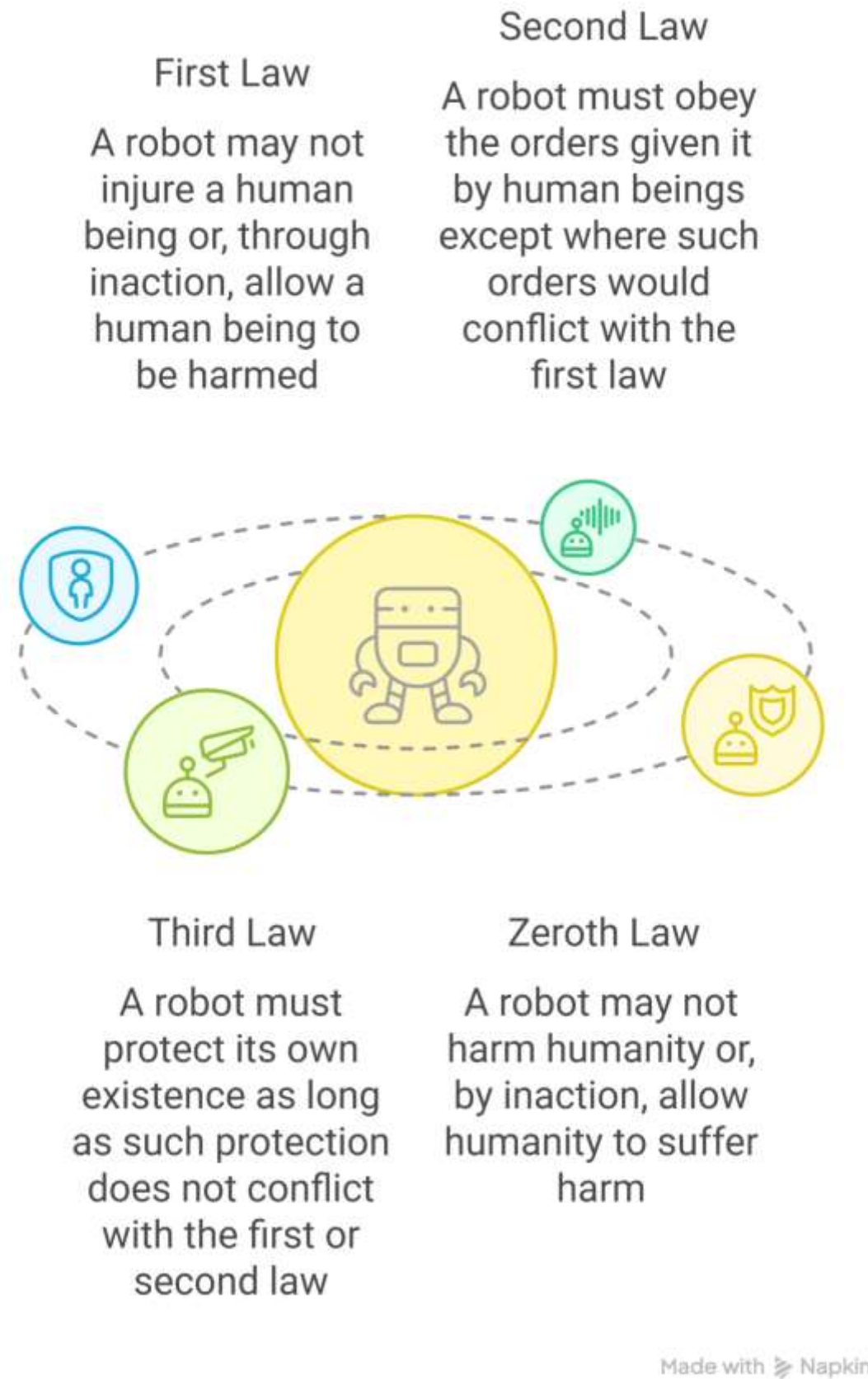
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Roboethics:

Roboethics focuses on **human responsibility when it comes to designing, building, using, and interacting with artificial beings (like robots).**

It covers a wide range of ethical questions regarding how humans should treat AI entities. Some of the critical areas include:

- **Designing ethical robots:** Ensuring that robots are built with features that align with human morals, including their capabilities and limitations.
- **Treatment of robots:** Addressing how humans should interact with AI entities, especially those designed to resemble humans.



2. Machine Ethics:

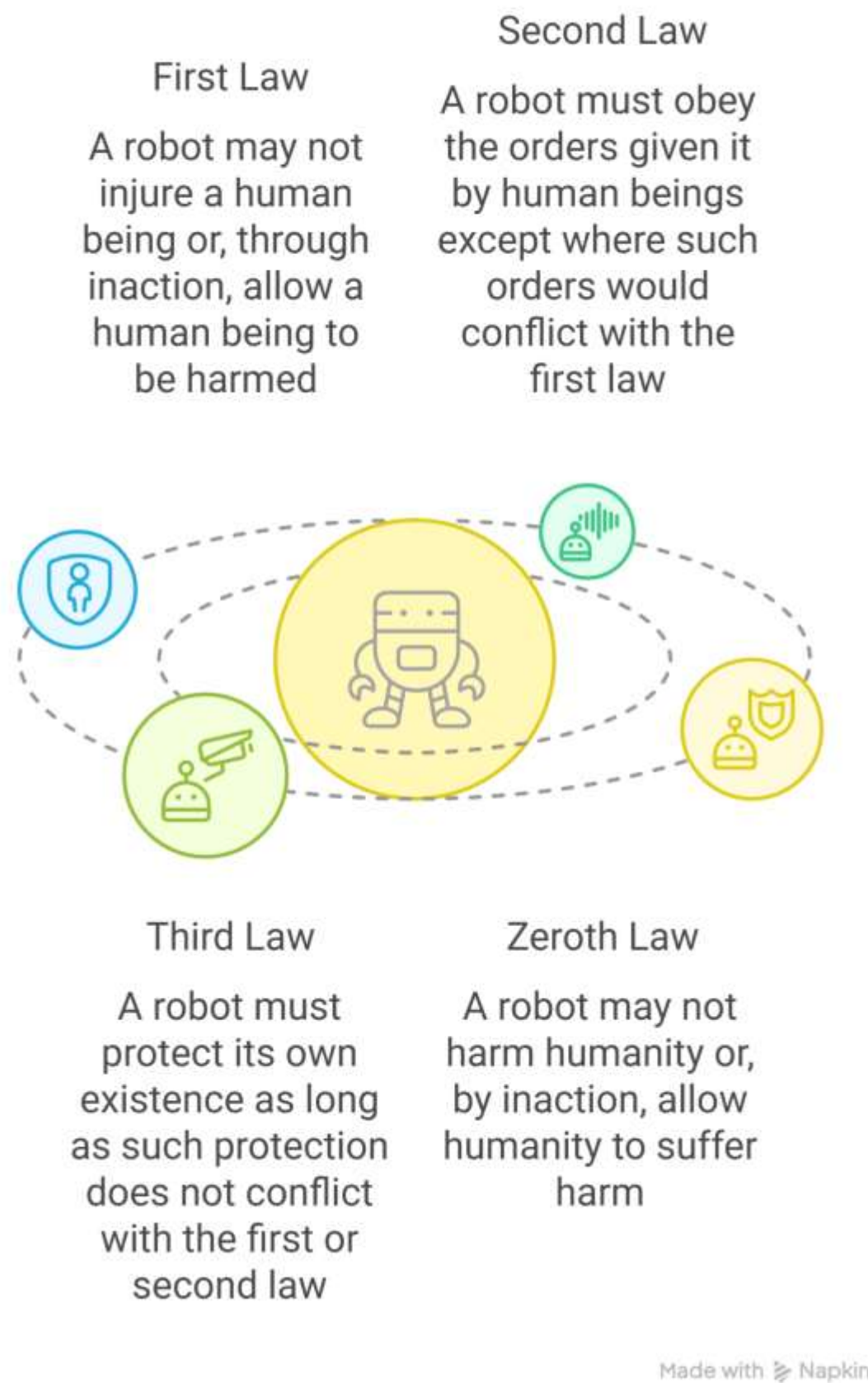
Machine ethics, on the other hand, **concerns the moral behavior of AI systems or Artificial Moral Agents (AMAs)**, especially those designed to interact with humans or make important decisions.

These systems need to make moral decisions independently, without direct human intervention.

Examples of AMAs include self-driving cars or AI systems used in healthcare. This includes:

Decision-making systems: AI algorithms that affect people's lives directly, like automated loan approvals, criminal sentencing, or hiring decisions.

Moral programming: Developing frameworks that allow AI systems to make ethical decisions based on societal norms, laws, and ethics.



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1. AI and Ethical Concerns

Key Ethical Concerns in AI

1. Can AI Make Human-like Decisions?

AI generates human-like outputs (e.g., loan approvals, healthcare decisions). Do AI systems understand the consequences of their actions like humans do?

2. Are AI Decisions Truly Human-like?

AI decisions are based on data and probabilities. But do they fully comprehend human context behind those decisions?

3. Justification of AI's Decisions

Is the reasoning behind AI decisions justifiable? Transparency is crucial for aligning AI decisions with ethical and legal frameworks.

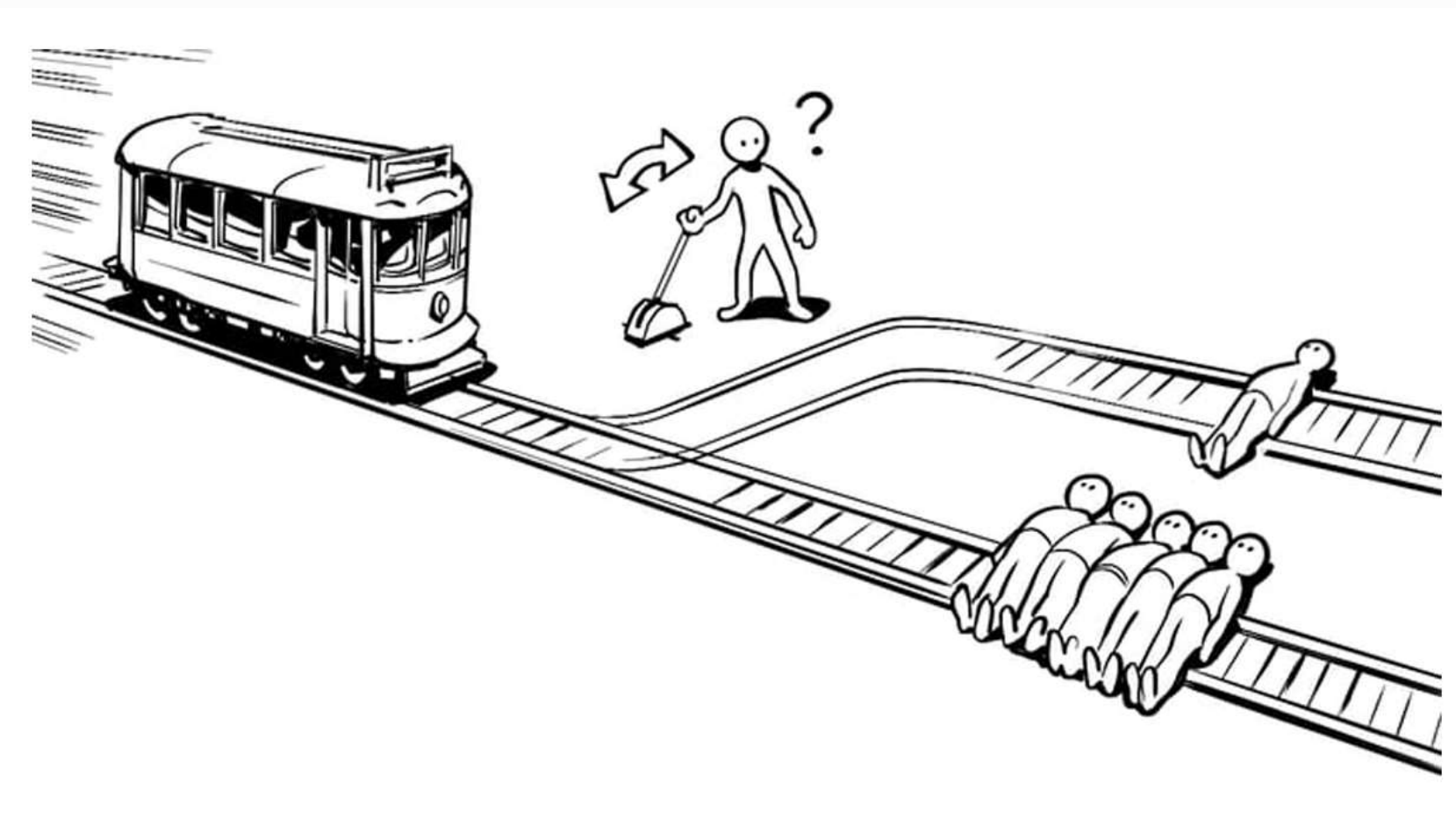
4. Fairness in AI Decisions

Are AI decisions fair and free from bias? AI systems should respect individual rights and avoid discrimination.

5. Trustworthiness of AI

Can we trust AI systems (e.g., self-driving cars) to make safe decisions? AI must be reliable, transparent, and accountable to earn trust.

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[I Asked ChatGPT for Its Take on the Trolley Problem | by Nolan Clarke | Medium](#)

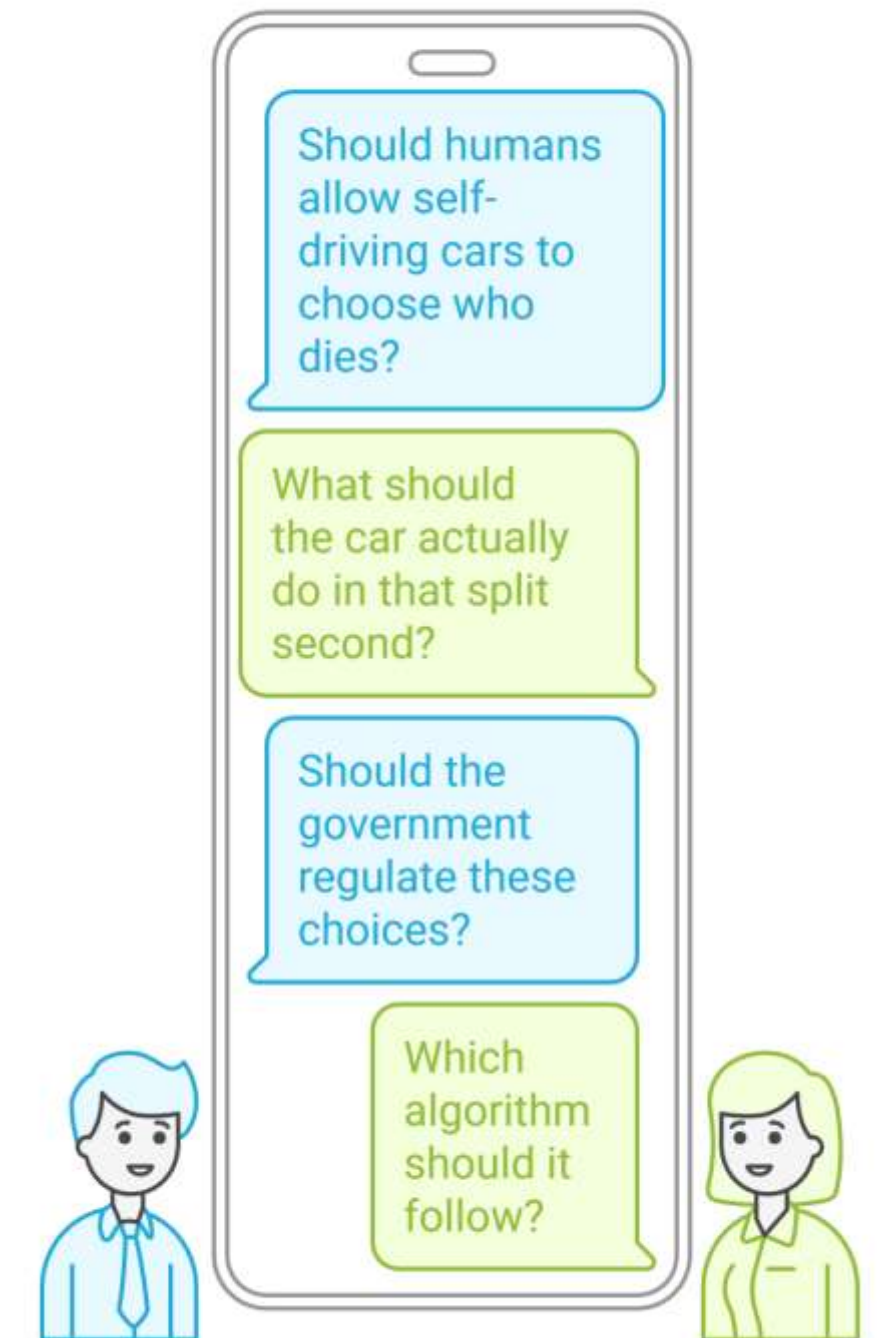
Utilitarianism: Choose the action that causes the least total harm.

Deontological Ethics: Certain actions are wrong even if the results are better.

- Robo Ethics is ethics *about* robots → guidelines for humans.
- Machine Ethics is ethics *inside* robots → guidelines for machines.



[Self-Driving Cars - Where are we now? | BlackboxMyCar](#)



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8.1.2 Is AI Dangerous? Will Robots Take Over the World?

AI's Potential for Harm

AI could be risky if specifically programmed for harm (e.g., autonomous weapons).
Autonomous drones and self-driving cars can become dangerous if misused or hacked.

Lack of Clear Accountability

Current laws don't cover AI's role in society, especially in terms of liability.

Questions arise like: Who is responsible for AI's actions? Can AI systems be held accountable?

Job Automation Risk

AI might replace many jobs, especially low-skilled, repetitive tasks. This could lead to unemployment for millions, particularly in industries like retail, hospitality, and warehouse work.

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8.1.2 Is AI Dangerous? Will Robots Take Over the World?

Biased Algorithms

AI systems are vulnerable to bias if trained with biased data. Biased AI can produce unfair results, impacting crucial decisions like hiring or loan approvals.

Privacy Concerns

AI systems collect large amounts of personal data, leading to privacy invasion. Facial recognition technology can track individuals without consent, creating privacy risks.

Unreliable Information

AI can be used to create fake content (e.g., deepfakes, fake news). Manipulated videos or images can spread misinformation or be used maliciously

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8.1.3 Ethics in AI

Ethical AI Definition

Ethical AI refers to AI systems designed to be **fair, just, and unbiased**.

It focuses on ensuring AI's actions align with moral principles.

Importance of Ethical AI

Ethical AI ensures the **technology benefits society while avoiding harm or unfair treatment**.

It guarantees AI decisions are transparent and responsible.

Bias in AI

AI systems can produce **biased results if trained on biased data**. Unchecked bias in AI can lead to unethical decisions, such as unfair hiring or loan approval.

Example of Unethical AI

Tay Chatbot: Microsoft's Tay learned negative and harmful behaviors after interacting with trolls on Twitter. The bot started spreading hate and offensive messages, forcing Microsoft to shut it down within 24 hours.

Features of Ethical AI

- a. Uses unbiased data for training.
- b. Follows the right learning model for the problem at hand.
- c. Continuously monitored to ensure its outputs are right and fair.

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8.1.4 AI and Bias

What is AI Bias?

AI bias occurs when an algorithm produces results that are **unfair** or prejudiced towards a particular group.

Example: If AI frequently misidentifies people of color in computer vision systems, it's biased.

Examples of AI Bias

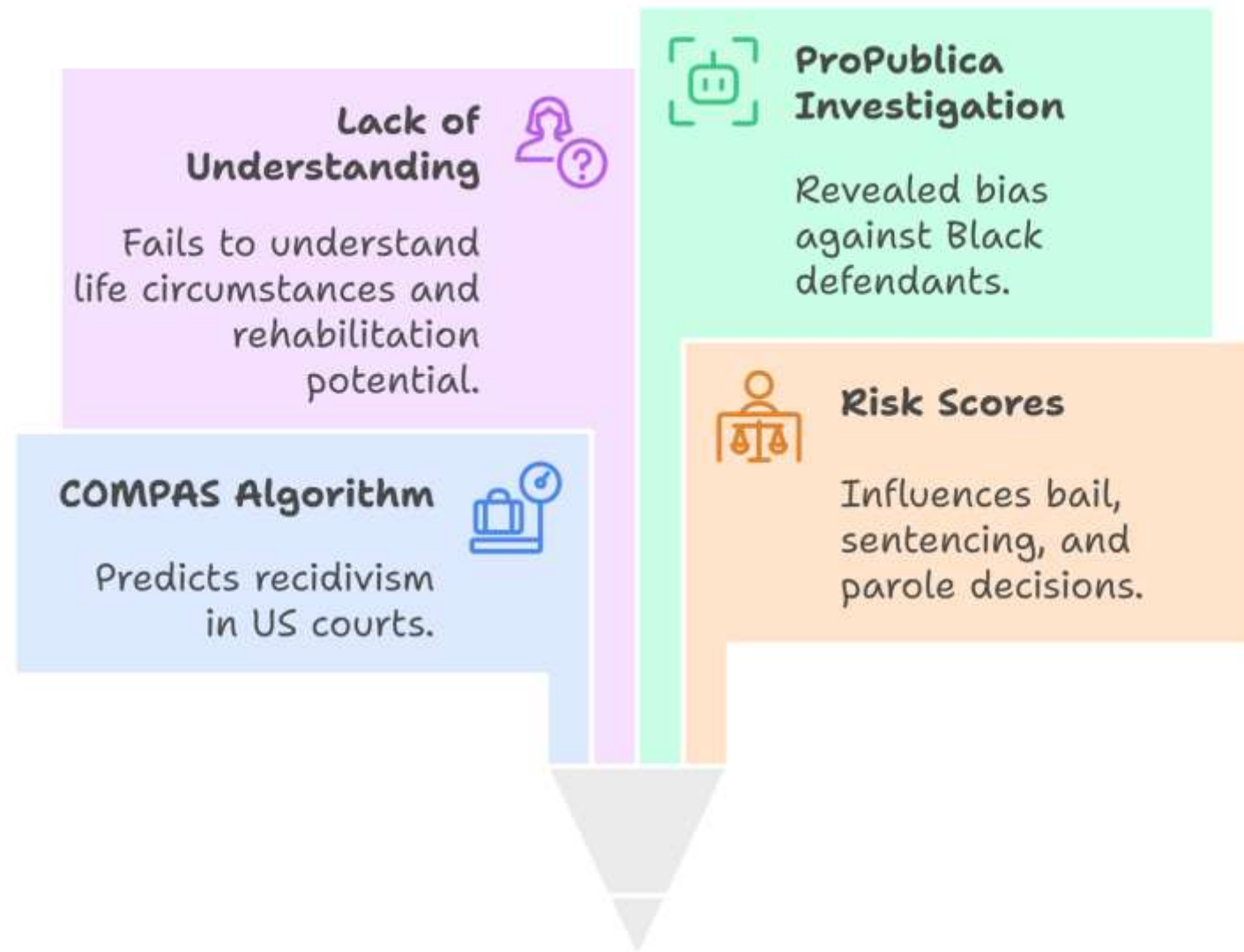
Doctors in Images: Search results for "doctor" mostly show white men.

Gender Bias in Professions: "Doctor" is often male, while "nurse" is female in many images and videos.

Voice Assistants: Virtual assistants like Siri and Alexa use female voices, reinforcing gender bias.

Recognition Failures: Computer vision struggles with recognizing people of different skin colors accurately.

COMPAS Recidivism Algorithm



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Amazon's Recruiting AI (2018)



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The Case of Harvey AI



Appearance of intelligence ≠ real comprehension; highlights the gap between human reasoning and AI prediction.



AI decisions may seem human-like but lack true human understanding or ethics.



Outputs appear intelligent but lack legal judgment, empathy, and accountability.



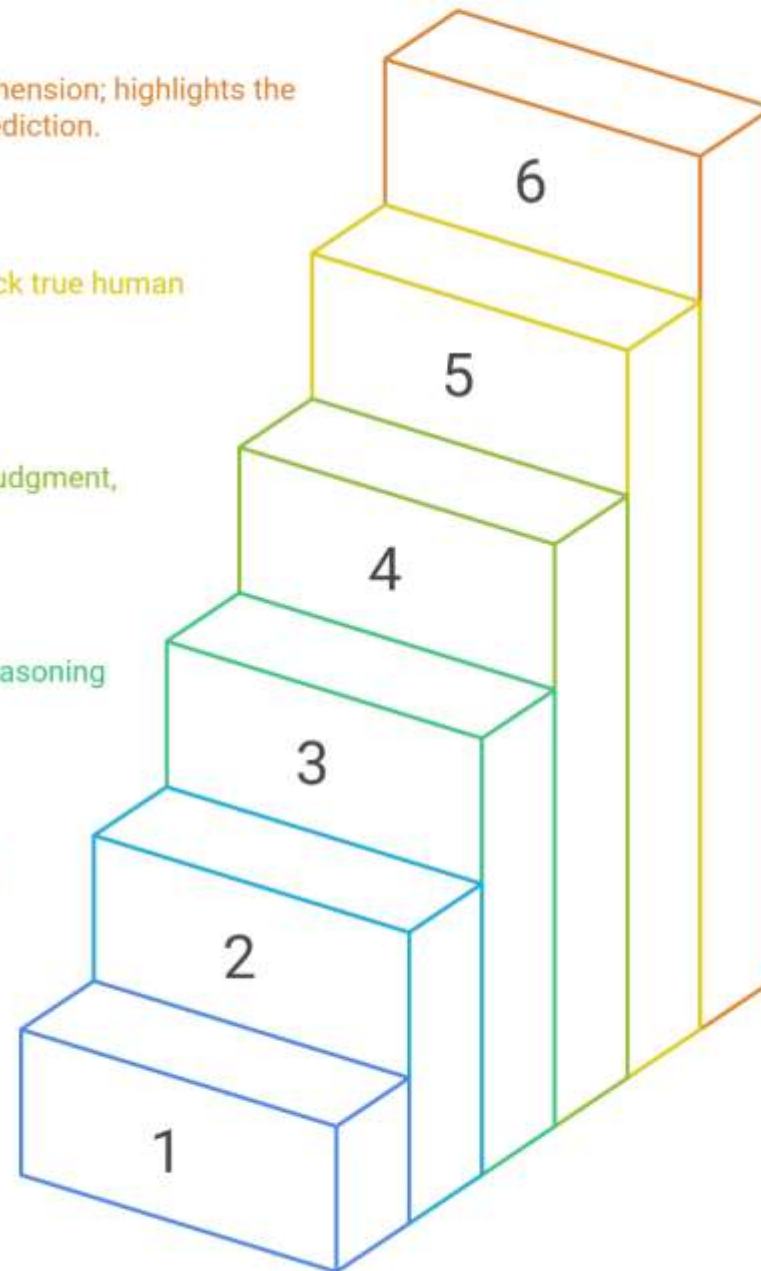
Critics view it as a "GPT wrapper," not a reasoning lawyer.



It predicts words based on data patterns, lacking true legal understanding.

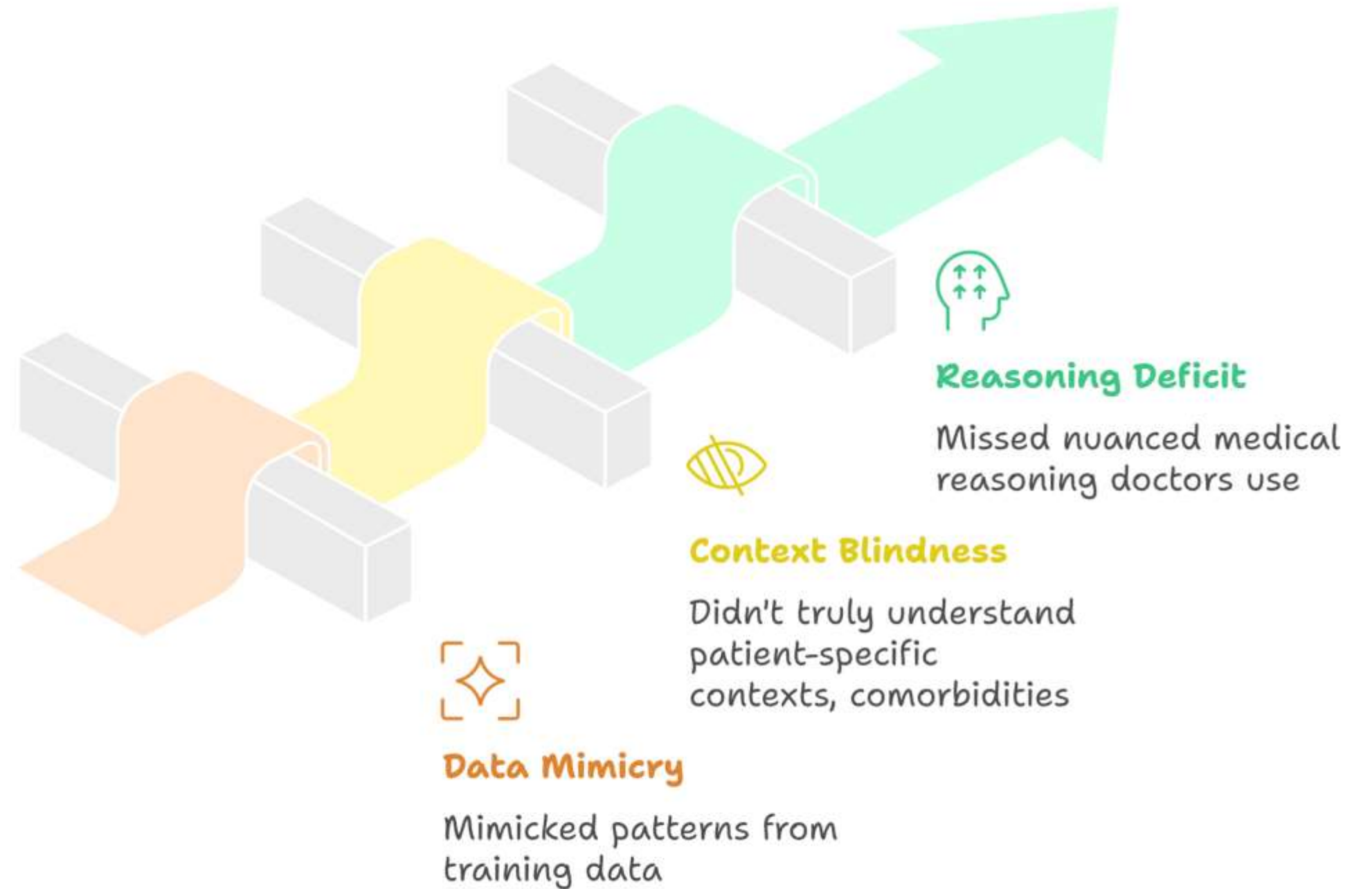


Harvey AI assists lawyers in drafting contracts and reviewing documents.



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IBM Watson for Oncology



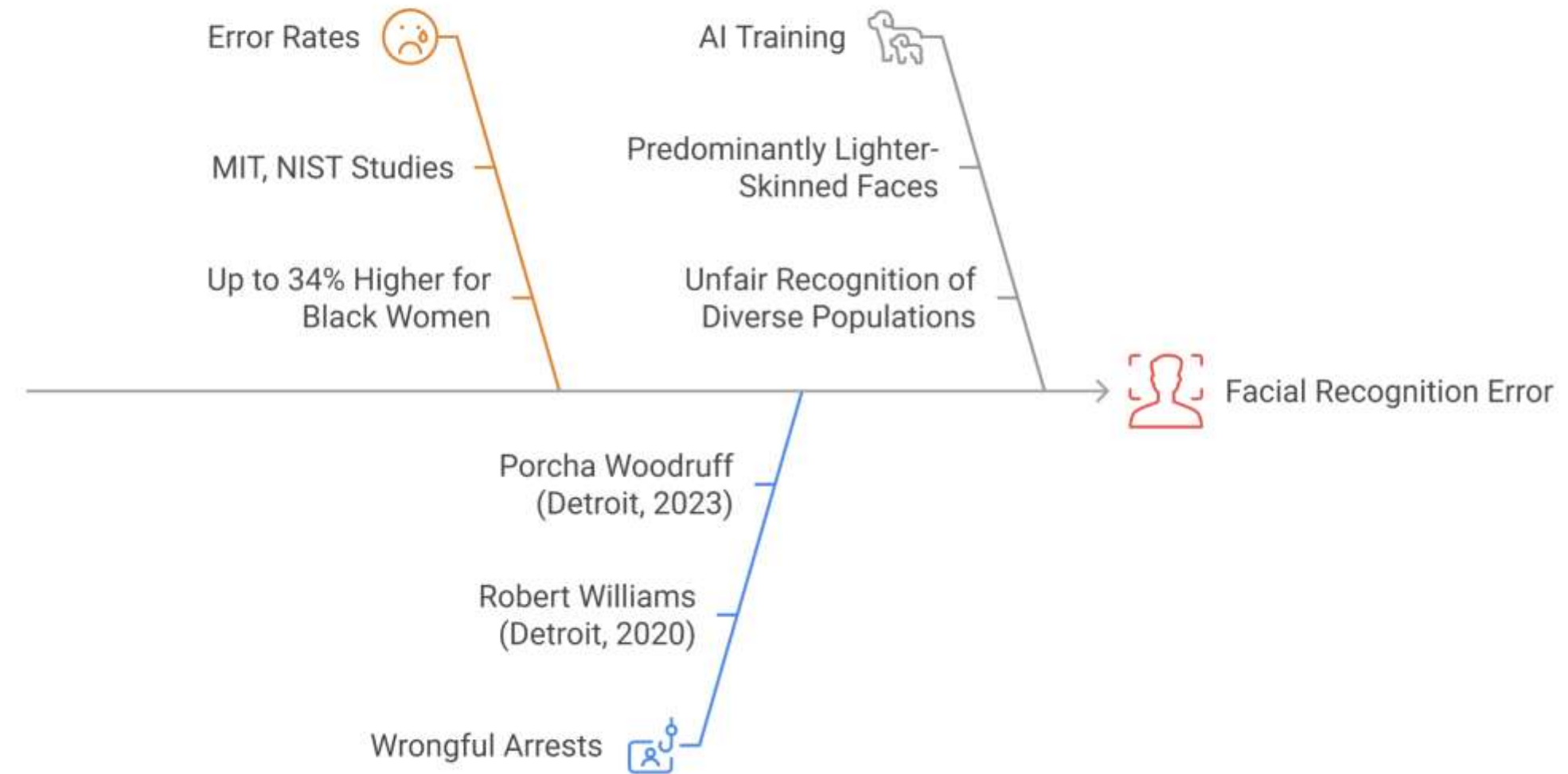
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Dutch Childcare Benefits Scandal



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Facial Recognition Error Rates and Consequences



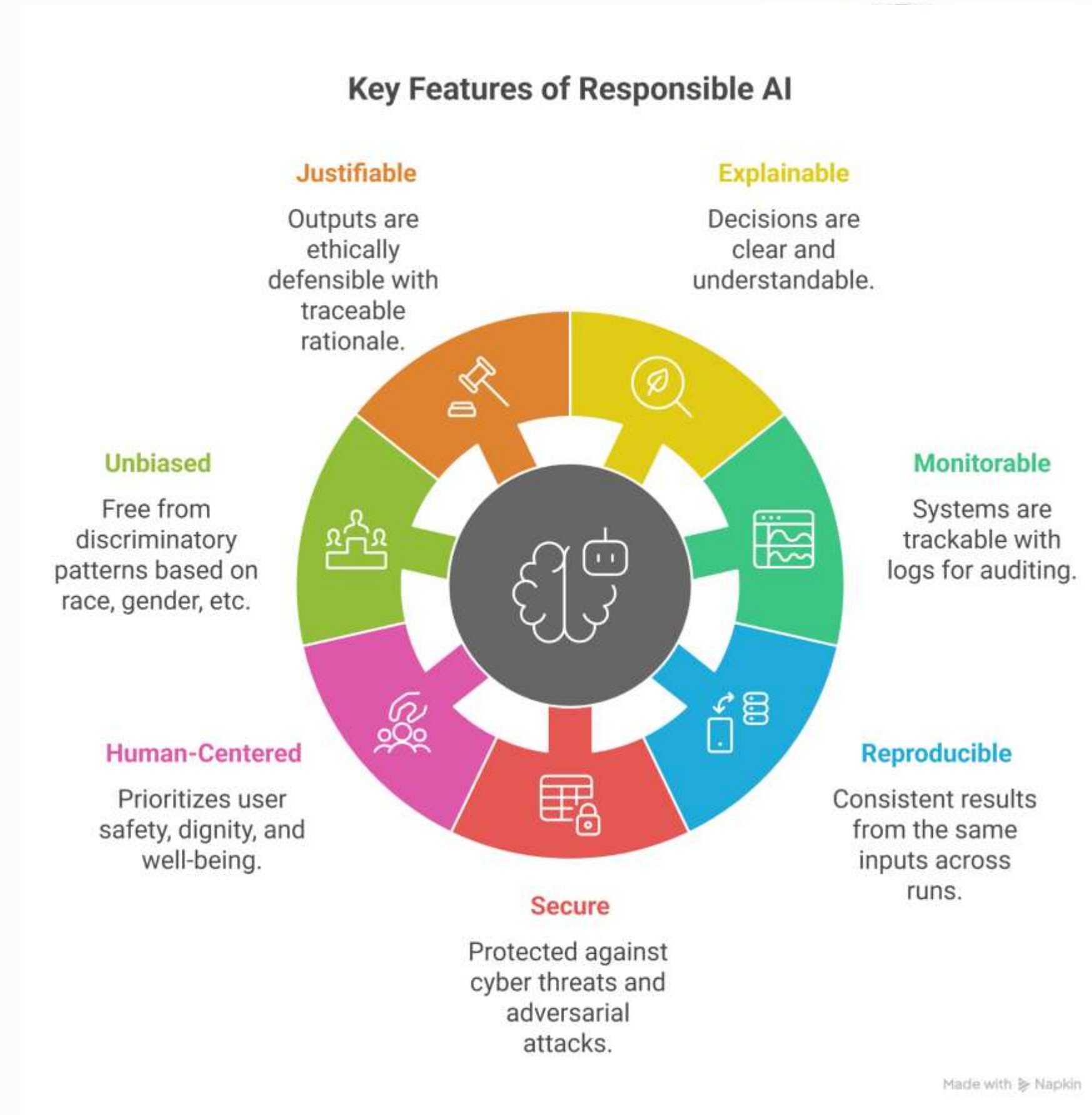
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8.1.1 Ethical Use of Artificial Intelligence

Challenges in Regulation

Formulating laws for AI is difficult due to rapid advancements in technology.

Existing laws (e.g., privacy) may not cover new AI technologies like voice assistants (e.g., Alexa, Siri), which can be exploited by malicious actors.



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8.1.5 Towards Ethical and Trustworthy AI

Company and Organizational Engagement:

Building Trust: Organizations are implementing:

Cultural programs to raise awareness. Risk assessments to identify potential AI issues. Third-party audits to ensure compliance with ethical standards.

Framework: AI systems should be monitored using a prevention, detection, and response framework (like anti-corruption measures).

Role of Rights and Activist Groups:

Human-Centric AI: AI systems must **prioritize human rights and be developed with social responsibility.**

Challenge: Civil rights groups should challenge AI practices and amplify the voices of those most affected by technology.

Questions to Ask:

Is the technology necessary? Is there an alternative? Does the benefit outweigh the harm?

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8.1.5 Towards Ethical and Trustworthy AI

Ensuring Data Privacy: **Data Protection:** Essential for ethical AI. **Encryption:** Ensures data is secure.

Secure Computation: Prevents AI developers from accessing private data.

Dataset Boundaries: AI should not learn beyond its provided dataset.

Prevent Reverse Engineering: Ensures data used in AI models is inaccessible to users.

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Diversify Your Team to Prevent Bias:

Diverse Teams: Forming teams with varied backgrounds (e.g., data scientists, business leaders, lawyers, sociologists, and ethicists) can help in identifying and preventing bias in AI systems.

Fairness Checks: Regular analysis of data and algorithms ensures that biases are spotted early.

Bias Mitigation Tools: Tools like Bias Analyzer help automate the detection of bias and analyze the effectiveness of mitigation actions.

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8.1.6 Why is Ethical AI Important?

Critical Areas of Application:

AI is used in medicine, law enforcement, recruitment, data privacy, military defense, and self-driving vehicles.

These systems must produce accurate, transparent, and understandable results, aligned with societal ethical standards.

Risks of Unethical AI:

Biased or incorrect outputs can:

Erode privacy (e.g., misusing personal data).

Lead to decisions that are impossible to understand. Cause harm without clear accountability for damages.

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8.1.6 Why is Ethical AI Important?

Importance of Fairness & Accuracy:

AI must be unbiased and accurate Without ethical design, AI systems can be misused, leading to disastrous consequences for humanity.

Examples of Unethical AI Outputs:

Medical Bias: An algorithm used in US hospitals to predict patients needing extra care favored white patients over black patients due to biased data regarding medical billing and facilities.

Hiring Bias: Amazon's AI hiring algorithm discriminated against women due to a data bias from previous male-dominated applications.

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8.1.6 Why is Ethical AI Important?

Efforts to Ensure Ethical AI:

IEEE Global Initiative and other committees focus on making AI systems more ethical.

Ethically Aligned Design (2016): A publication providing guidelines for ethical AI solutions.

Guiding Principles for Trustworthy AI (European Commission High-Level Expert Group):

Support Human Autonomy: AI should enhance human decision-making. **Robust & Safe:** AI must be technically sound with a fallback plan for failure. **Data Privacy:** Protect user data and maintain data quality.

Transparency: AI should avoid biased data and be transparent about its models.

Fairness: Train AI systems with non-discriminatory, diverse data to avoid bias. **Benefiting Everyone:** AI should promote societal and environmental wellbeing. **Accountability:** AI must be responsible for accurate, unbiased outcomes.

Data Security: Ensure confidentiality, transparency, and safety in data management.

Accountability for Decisions: Ensure organizational responsibility for AI decisions.

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2. AIasaService(AIaaS(

1. Factors Triggering Growth of AIaaS

- **Cloud Platforms:** Availability of affordable cloud platforms for enterprise data management.
- **Improved Data Storage:** Cost-effective and reliable data storage technologies.
- **Data Generation:** Streaming devices and IoT generate massive data that can be analyzed for competitive advantage.
- **Automation Tools:** Semi or fully automated data management and analytics tools simplify business analysis.

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8.2 AI as a Service (AIaaS)

8.2.2 The Growth of AIaaS

Accessibility: Cloud services make AI available to more companies, reducing the need for massive investment in talent and resources.

Key Benefits:

Focus on Core Business: Companies can focus on their core business instead of learning AI.

Cost Reduction: AIaaS lowers operational costs and avoids expensive infrastructure.

Minimized Investment Risk: No huge upfront costs or resource commitments.

Quick Deployment: AIaaS offers fast solutions with pre-built algorithms.

Data Insights: Gain competitive edge by analyzing trends and improving data usage.

Scalability: Companies can start small and scale AI projects as needed.

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2. AIaaS(AIaaS)

3. Challenges of AIaaS

- 1.**Security:** Sharing data with third-party vendors can raise security concerns.
- 2.**Reliance on Third Parties:** Dependence on vendors for services can lead to delays.
- 3.**Reduced Transparency:** Lack of insight into the internal workings of AI systems.
- 4.**Data Governance:** Industry-specific data policies may restrict the use of third-party AIaaS.

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8.2 AI as a Service (AIaaS)

8.2.4 Vendors of AIaaS Major AIaaS Providers:

Amazon Web Services (AWS): Offers machine learning services like bots, APIs, and frameworks.

Microsoft Azure: Provides AI and ML solutions, including cognitive services and bot services.

Google Cloud Platform (GCP): Helps data scientists and developers create and deploy machine learning projects.

Other AIaaS Providers:

Monkey Learn: Simplifies text analysis using no-code tools.

IBM Watson: Offers a suite of AI tools for building virtual assistants and text analysis.

Microsoft Azure: Provides cognitive services for computer vision and text extraction. **Google Cloud ML:** Facilitates data scientists in creating and deploying ML models.

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8.2 AI as a Service (AIaaS)

8.2.5 Market Growth

SaaS Market: Valued at \$133 billion.

AI Platform-as-a-Service (AIPaaS): Expected to grow from \$11 billion in 2023 to \$88.5 billion by 2025.

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4. Recent Trends in AI

1. Collaborative Systems Machine-Human Collaboration:

- AI and humans work together to complement each other's strengths.
 -
- **Examples:**
 - Foldit: AI and humans collaborate to fold proteins.
 - AI in Chess: Amateurs and AI beat grandmasters. AI in Business:
 - AI elected to company boards. Military: Drones and humans
 - working together.
 - Exoskeletons: Power suits assist soldiers in the field.
- **Challenges:**
 - Over-reliance on AI could lead to job loss and skill degradation.

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8.4 Recent Trends in AI

8.4.2 Machines Assisting Humans

- **AI Helps Humans Enhance Abilities:**
 - AI boosts creativity, decision-making, and interactions.
- **Examples:**
 - SEB Bank's AI Assistant: Helps customers with tasks and understands tone.
 - AI improves efficiency in various sectors like healthcare, customer service, and homes.

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8.4 Recent Trends in AI

8.4.3 Algorithmic Game Theory Game Theory with AI:

AI helps machines play games ethically using predefined rules.

Types of Games:

Cooperative vs. Non-cooperative: Alliances or no alliances.

Symmetric vs. Asymmetric: Same vs. different goals. **Perfect vs. Imperfect Info:**

Full visibility vs. hidden moves. **Zero-sum vs. Non-zero sum:** One loses, one wins vs. all gain.

Key Concepts:

Nash Equilibrium: No player can improve their situation by changing strategies.

Inverse Game Theory: Designing games based on players' strategies.

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8.4 Recent Trends in AI

8.4.4 Multi-Agent Reinforcement Learning (MARL)

Multiple Agents Learning:

Multiple AI agents interact and learn together.

Example:

Self-driving cars: Cars learning to manage traffic flow.

Challenges:

Increased complexity with more agents interacting.

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9.2 Internet of Things

IoT is a system of interrelated computing devices, machines, objects, animals, or people with unique identifiers that can transfer data over a network without human-to-human or human-to-computer interaction.

Examples of IoT Devices:

People: Heart monitor implants. Animals: Biochip transponders.

Automobiles: Sensors for monitoring tyre pressure.

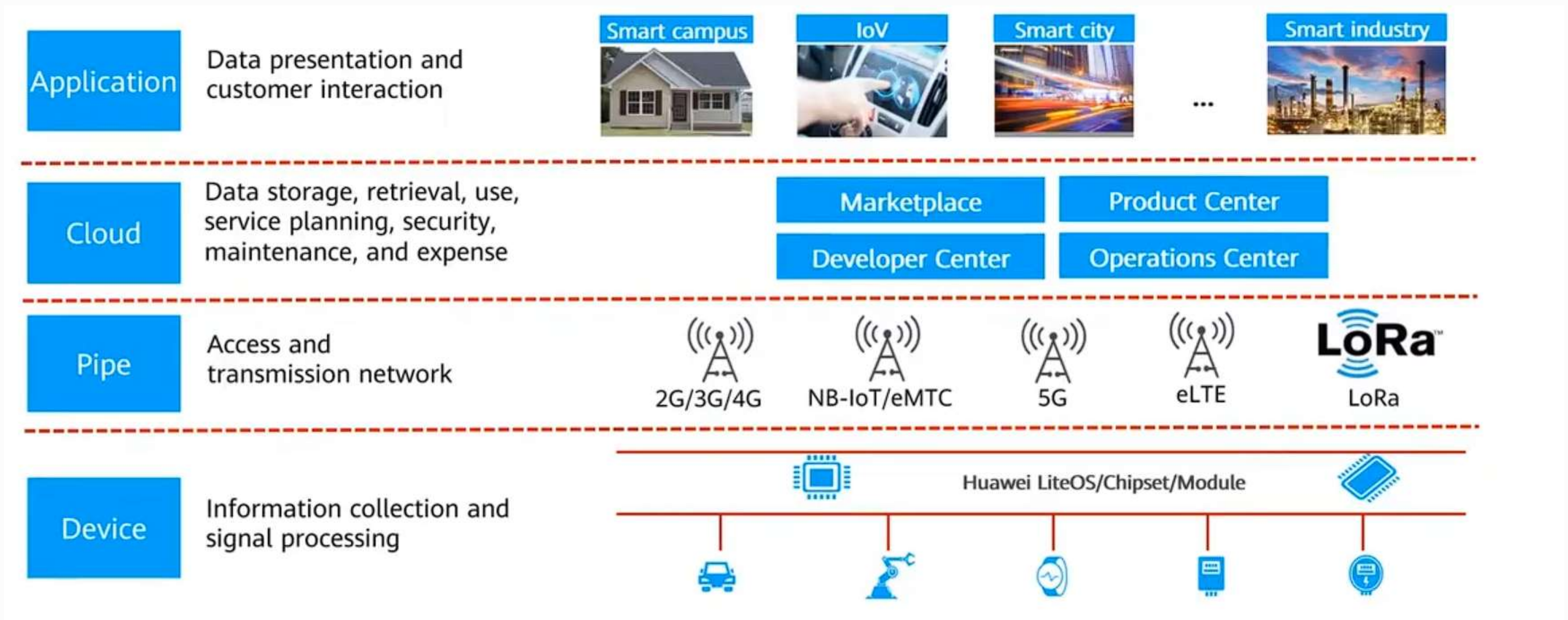
Objects: Any device assigned an IP address to transfer data.

Applications of IoT:

Precision Agriculture: Monitoring crops, soil, and environment.

Building Management: Automating lighting, heating, and security systems. Healthcare: Remote patient monitoring with wearable devices.

Energy: Smart grids and energy management systems. Transportation: Traffic management, vehicle monitoring, etc.



[What is AIoT](#) | [Basic Concept of AIoT](#) | [Overview](#) | [AI and IoT](#)

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9.2.1 Examples of Applications of IoT

Smart Car Navigation:

Car suggests the best route, updates on traffic, and sends notifications if delayed.

Smart Home Automation:

Alarm rings, and coffee machine starts brewing automatically.

Wearable Fitness Trackers:

Smartwatch tracks activity and productivity, providing insights.

Self-Driving Cars:

Equipped with sensors for detecting objects and ensuring safe navigation.

Smart Football:

Tracks distance and speed, records stats for training.

Smart Refrigerator:

Notifies when groceries run out or when food is expired.

Smart Air Conditioner:

Controlled remotely via smartphone to set the desired temperature before arrival

Smart Cities:
IoT helps in reducing traffic congestion, managing resources efficiently, and reducing pollution and crime.

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9.2.2 IoT Products

Amazon Echo (Alexa):

Voice assistant for smart homes.

Can perform various tasks like playing music, providing weather reports, or ordering an Uber.

Fitbit One (Wearables):

Tracks physical activities (steps, calories, sleep). Syncs with smartphones and computers to monitor progress.

Barcelona - Smart City:

Implemented IoT solutions for smart parking and environmental improvements. One of the leading smart cities globally.

Popular IoT Platforms:

Amazon Web Services (AWS) Microsoft Azure IBM Watson Cisco IoT Cloud Connect Salesforce IoT Cloud Oracle Integrated Cloud GE Predix

AT&T - Connected Car:

- Added 1.3 million connected cars to its network in 2016. Brings the total to 9.5 million connected cars.

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9.2.3 Challenges in IoT

Security:

Privacy and data sharing concerns. Hackers target connected devices (e.g., cars, homes). Lack of encryption in communication can cause security vulnerabilities.

Scalability:

Huge volumes of data generated need big data analytics and cloud storage. Devices must be scalable to process and store ever-increasing data.

Interoperability:

Lack of common standards for IoT devices. Difficulty integrating with legacy systems.

Lack of Standardization:

Standardization issues hinder IoT device compatibility. Prevents achieving truly connected systems.

Lack of Government Support:

Regulatory bodies (e.g., FDA) lack IoT-specific safety and security regulations. Limited governmental oversight.

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9.2.3 Challenges in IoT

IoT Malware and Ransomware:

IoT devices vulnerable to malware and ransomware attacks. Hackers can take control of devices and demand ransom.

Connectivity:

Poor internet connectivity affects IoT device performance. IoT devices require stable connections for effective monitoring and control.

Privacy Policies:

Lack of clear regulations on what data is private or public. Example: IoT toys collecting children's data without parental consent.

Bandwidth Strain:

Growing number of IoT devices creates strain on network bandwidth. Increased data flow demands higher bandwidth and powerful servers.

IoT Skills Gap:

Shortage of trained professionals in IoT network management. Difficulty in training new staff due to rapid IoT development.

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9.2.4 Sensors

Sensor Type	Description	Example
Oxygen Sensor	Detects the gasoline/oxygen ratio	Detects Car emission control system
Motion Sensor	interruption in energy flow	Home security lights, automatic doors
Vision and Imaging Sensor	Detects objects or colors in view Measures	Cameras for object detection Industrial
Temperature Sensor	thermal parameters	processes for gases, liquids, and
Radiation Sensor	Detects alpha, beta, or gamma	solids Surveys and sample counting
Proximity Sensor	Detects presence of objects without contact	Manufacturing operations Control
Pressure Sensor	Detects force per unit area in gases/liquids	and display devices Control
Position Sensor	Senses position of objects like valves or	systems for machinery
Photoelectric Sensor	doors Detects objects and can sense color	Manufacturing and material handling

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9.2.4 Sensors

Sensor Type	Description	Example
Particle Sensor	Detects airborne particulates	Bin and baghouse monitoring
Motion Sensor	Senses movement or stoppage	Conveyor detection, stalling bearings
Metal Detector	Senses the presence of metal	Security checks
Level Sensor	Determines height of substances in tanks	Control and display systems
Leak Sensor	Detects unwanted discharge of liquids/ gases	Monitoring gas leaks
Humidity Sensor	Measures water content in the air	Control systems for temperature regulation
Gas and Chemical Sensor	Detects gases/chemicals	Environmental monitoring
Force Sensor	Measures forces like weight or torque	Control systems for load measurement
Flow Sensor	Measures movement of gases/liquids/solids	Industrial processing systems
Flaw Sensor	Detects inconsistencies on surfaces	Manufacturing processes like welding

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9.2.4 Sensors

Sensor Type	Description	Example
Flame Detector	Senses fire presence and quality Detects	Combustion control systems
Electrical Sensor	electrical parameters like current and voltage	Monitoring electrical systems
Contact Sensor	Detects physical touch or contact Does not	Alarm systems for doors/windows
Non-contact Sensor	require physical touch	Radar guns for speed detection
Speed Sensor	Measures speed of an object/ vehicle	Vehicle monitoring
Ultrasonic Sensor	Measures distance using high-frequency sound waves	Object distance measurement in robotics

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3. Artificial Intelligence of Things (AIoT)

- AIoT combines Artificial Intelligence (AI) with the Internet of Things (IoT) to enhance efficiency, improve decision-making, and enable better data management. AI simulates
- human intelligence, while IoT connects devices to transfer data over networks without human intervention. AIoT improves IoT by adding machine learning algorithms to enhance
- decision-making.

1.● How Does AIoT Work?

- AI is embedded in IoT devices and connected through IoT networks. Devices collect data,
- which is analyzed using AI techniques to improve productivity and efficiency. Edge AI reduces bandwidth needs and minimizes delays in data processing.

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9.3 Artificial Intelligence of Things (AIoT)

9.3.3 Applications and Examples of AIoT

AIoT is widely used in various fields:

Smart Cities: Enhances operational efficiency, improves traffic management, and optimizes resource use.

Smart Retail: Uses smart cameras to track customers and optimize store operations.

Smart Homes: Devices analyze user habits to provide customized support.

Manufacturing: Monitors equipment and detects malfunctions.

Autonomous Vehicles: Uses real-time data to make driving decisions.

Healthcare: Devices monitor patient data, enabling remote care and accurate diagnoses. **Smart**

Thermostats: AI-powered devices adjust temperature based on user preferences and schedules.

Fleet Management: Monitors vehicles, tracks fuel usage, and identifies driver behavior.

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9.3 Artificial Intelligence of Things (AIoT)

9.3.4 Benefits and Challenges of AIoT Benefits:

Increases operational efficiency by analyzing and adjusting system operations in real-time.

Enables on-the-fly decision-making with real-time data analysis.

Reduces workload and saves time by automating data analysis. Scalable systems that connect more devices as needed.

Better risk management by predicting potential problems in advance. Improves products and services through real-time data analysis.

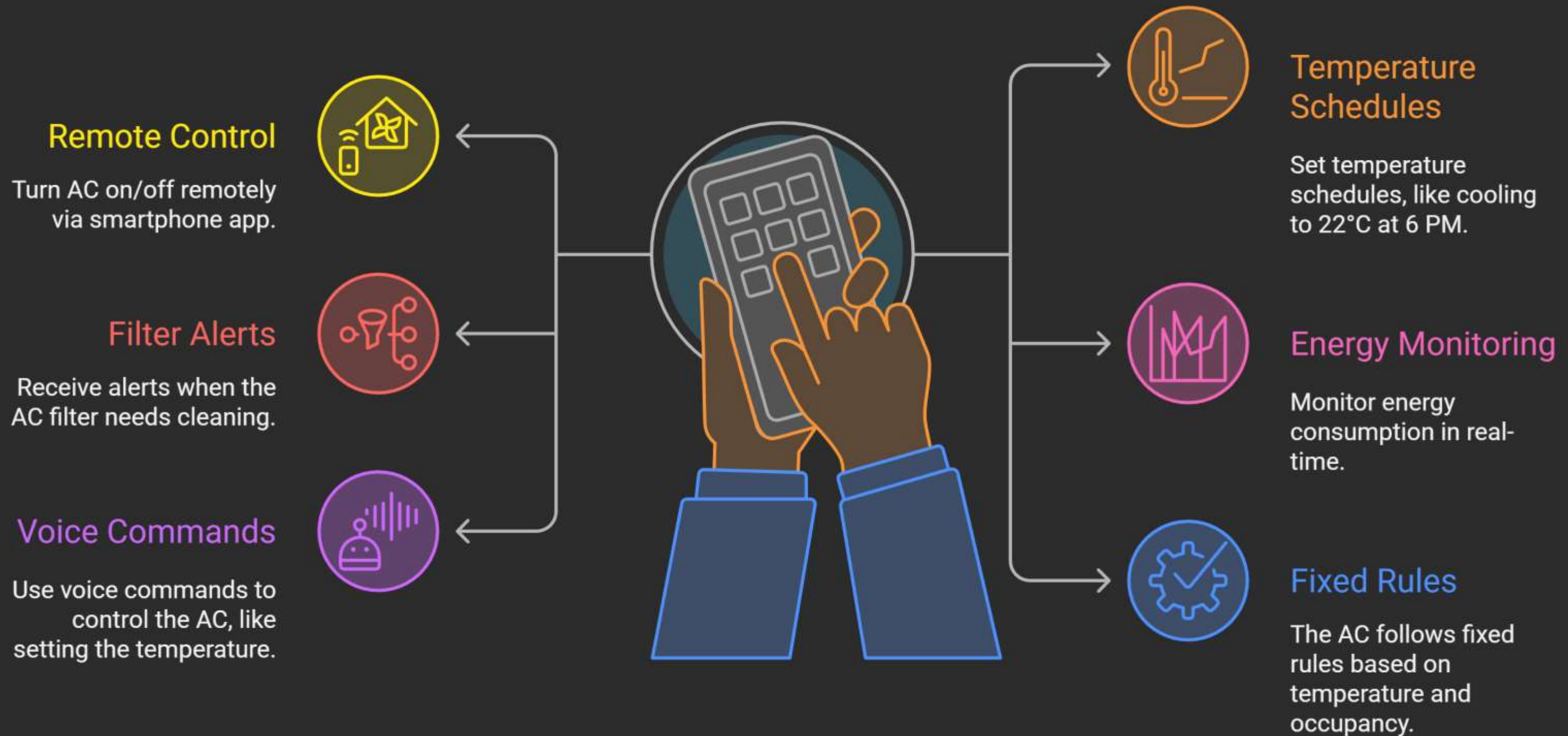
Challenges:

Security: Protection of sensitive data is a significant concern.

Scalability: Managing large volumes of data from multiple devices. **Interoperability:** Lack of standardization among different IoT devices. **Privacy:** Handling user data and ensuring privacy compliance.

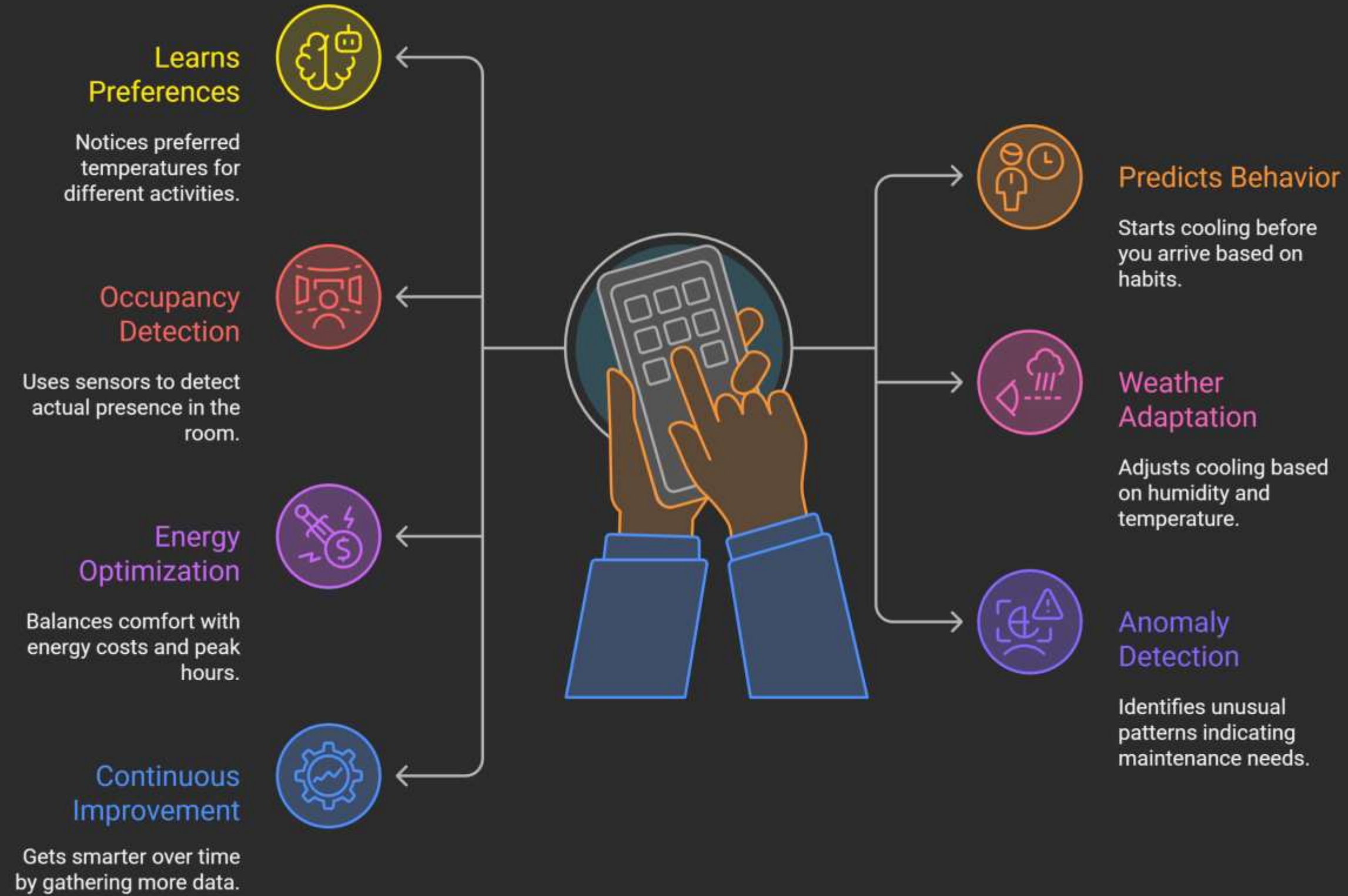
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Smart AC - IoT



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Smart AC - AIoT



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9.1 Expert System

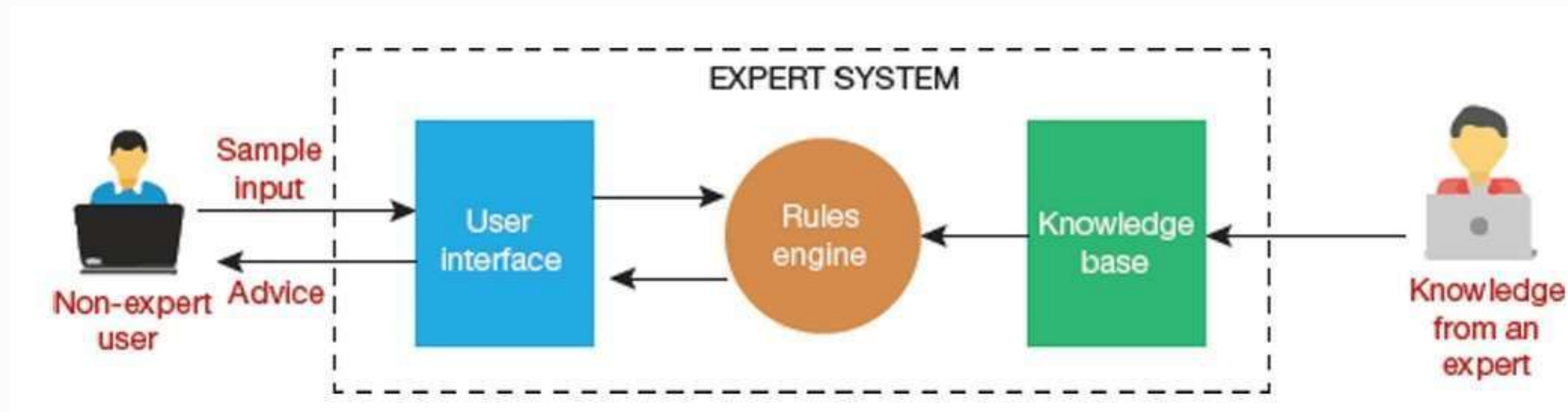
A computer program designed to solve complex problems with human-like expertise.

Key Components:

a. User Interface: Allows interaction with the system. b. Knowledge Base (KB): Stores expert knowledge for decision-making. c. Inference Engine: Applies reasoning rules to extract knowledge and provide decisions.

How it Works:

Expert systems analyze user queries. Extract knowledge from the knowledge base. Use reasoning and inference rules to generate results.



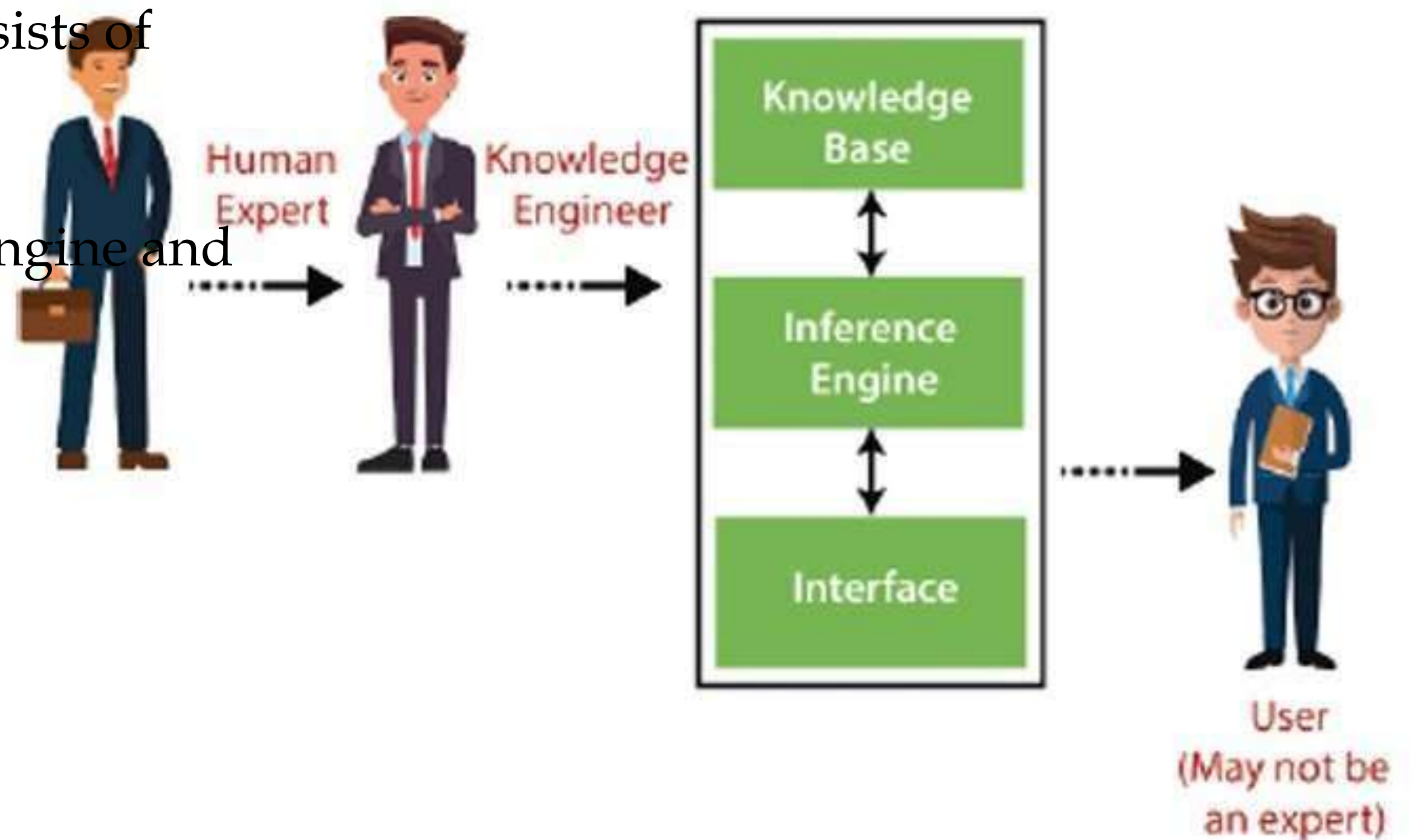
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9.1.3 Components of an Expert System

An expert system mainly consists of three

components –

1. User interface, 2. Inference engine and
3. Knowledge base



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9.1.3 Components of an Expert System

1. User Interface

Role: Allows users to interact with the system.

Function: Accepts user queries in natural language and provides understandable results.

Key Features:

Helps users accomplish tasks quickly.

Customizable for user requirements. Efficiently handles user input.

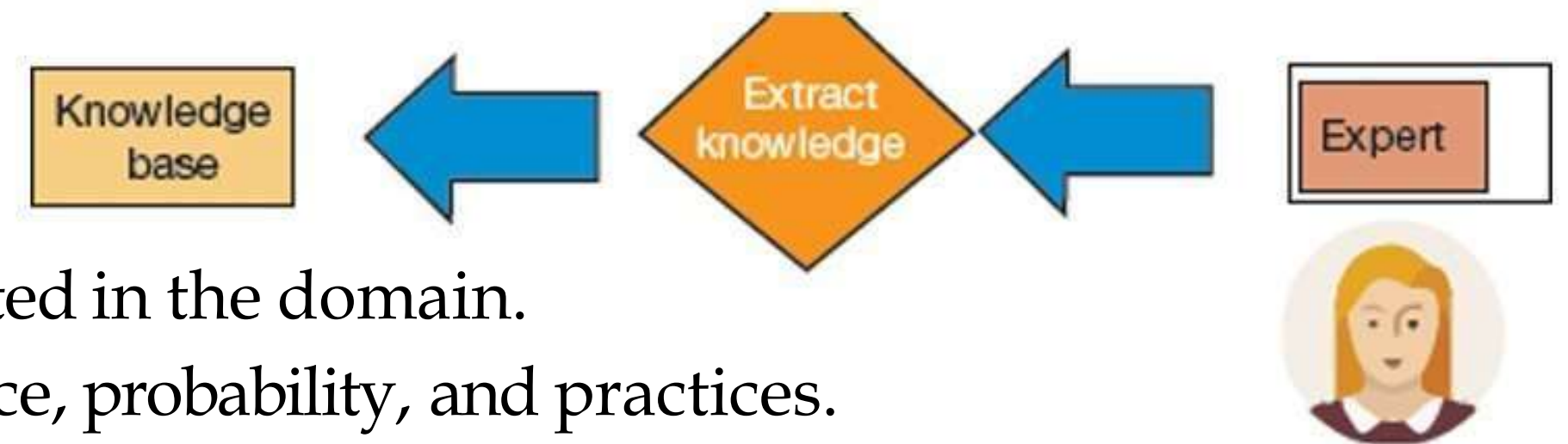
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9.1.3 Components of an Expert System

3. Knowledge Base

- **Role:** Stores domain-specific knowledge.
- **Types of Knowledge:**
 - **Factual Knowledge:** Based on facts accepted in the domain.
 - **Heuristic Knowledge:** Based on experience, probability, and practices.
- **Representation:** Knowledge is typically represented using If-Else Rules.



Case Study: MYCIN (Expert System)

- **Step 1:** Knowledge base about bacterial infections is created.
- **Step 2:** The system is updated with new information.
- **Step 3:** The doctor inputs details of a patient (e.g., symptoms, medical history).
- **Step 4:** The system applies If-Then rules using the inference engine.
- **Step 5:** Output is provided through the user interface, guiding diagnosis and treatment.

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9.1.3 Components of an Expert System

Inference Engine (Rules Engine) Strategies for Knowledge Acquisition:

Forward Chaining:

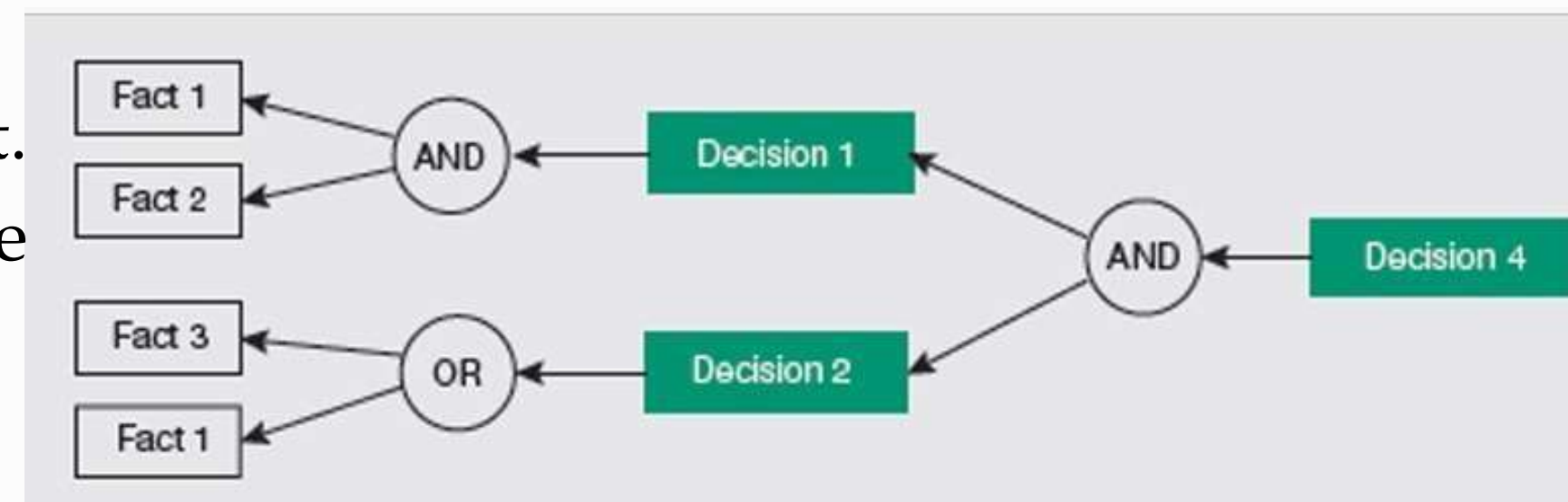
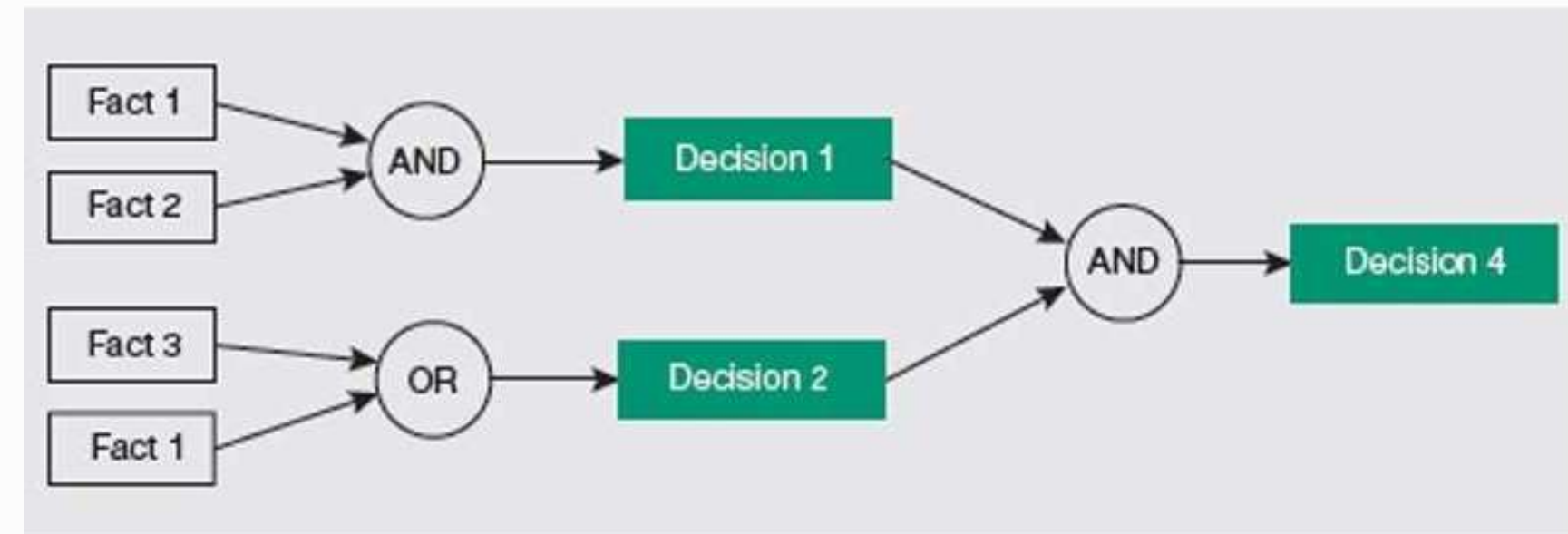
How it works: Starts with known facts and rules to predict what can happen next.

Example: Predicting stock prices after interest rate changes.

Backward Chaining:

How it works: Starts from the goal and works backward to prove the facts that lead to it.

Example: Diagnosing blood cancer by tracing the cause from the symptoms.



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9.1.3 Components of an Expert System

Inference Engine (Rules Engine)

The Inference Engine is the core processing unit of an expert system, often referred to as its "brain." It processes knowledge from the knowledge base and draws conclusions based on inference rules.

Types of Inference Engines:

1. Deterministic Inference Engine:

How it works: Uses facts and rules to draw conclusions. **Output:**

Conclusions are assumed to be true based on the given data.

2. Probabilistic Inference Engine:

How it works: Deals with uncertainty and draws conclusions based on probability.

Output: Conclusions have some level of uncertainty attached.

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9.1.3 Components of an Expert System

Inference Engine (Rules Engine) Functions of Inference Engine:

Efficient Processing: Applies rules and procedures to generate correct solutions. **Conflict Resolution:** If multiple rules apply, the engine selects the best rule based on the current context. **Explanation and Debugging:** Can explain how conclusions are reached, ensuring transparency.

○

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9.1.1 Popular Examples of Expert Systems

DENDRAL:

A chemical analysis system. Used in organic chemistry to identify unknown molecules by analyzing mass spectra.

MYCIN:

Designed to diagnose blood clotting diseases and bacterial infections. Recommends antibiotics and drugs based on the patient's condition.

PXDE

S: Determines the type and level of lung cancer using images of the upper body.

CaDeT:

Diagnostic support system to detect cancer at early stages.

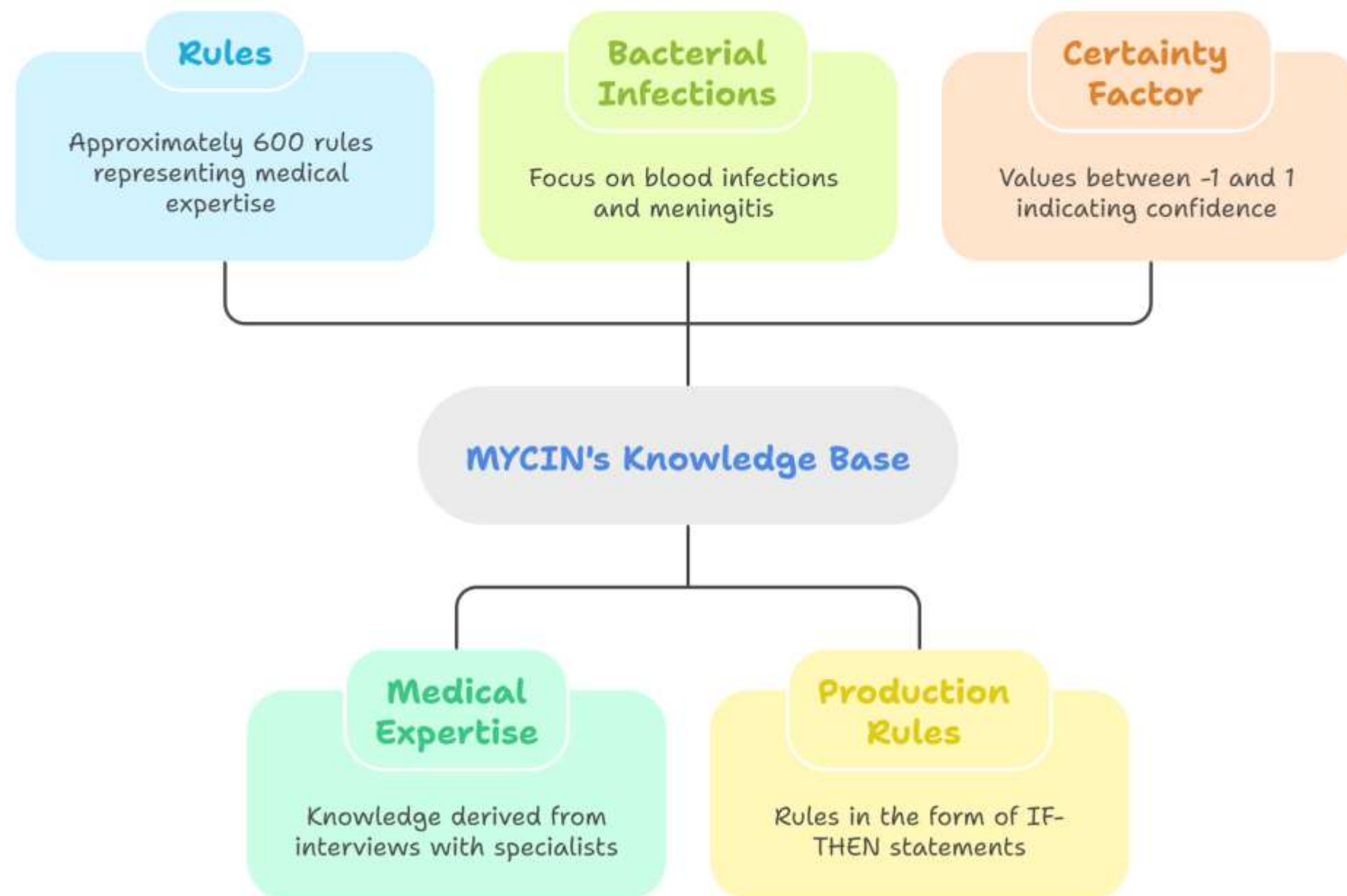
R1/XCON:

Customizes computer systems based on user requirements.

DXplain:

Clinical support system that suggests diseases based on doctor's findings.

MYCIN's Knowledge Base

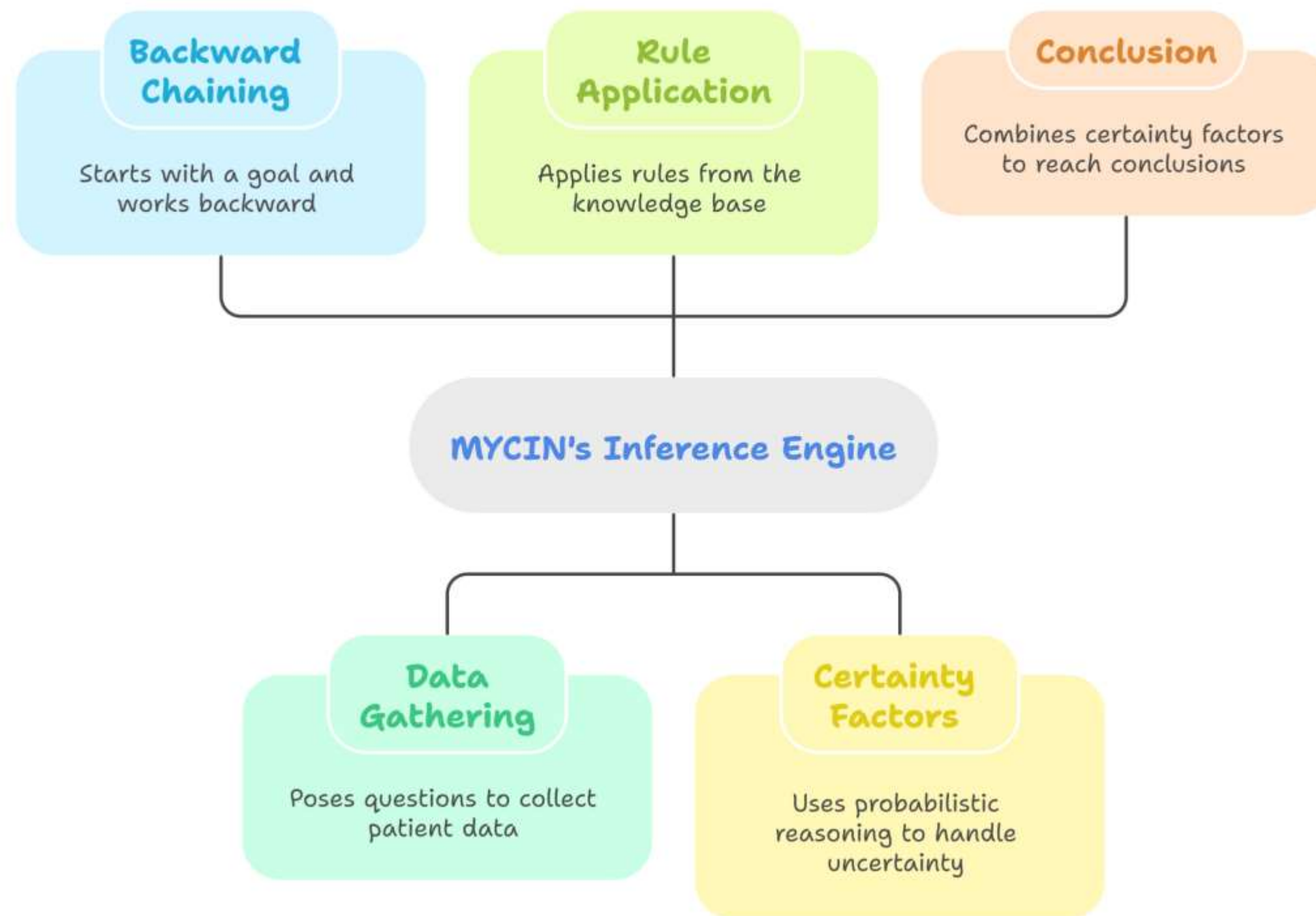


Made with Napkin

The knowledge base encoded relationships between symptoms, patient history, laboratory findings, and likely infectious organisms, mimicking how expert physicians reason about diagnoses

IF the infection is primary-bacteremia
AND the site of the culture is blood
AND the suspected entry point is the gastrointestinal tract
THEN there is evidence (0.7) that the organism is Bacteroides

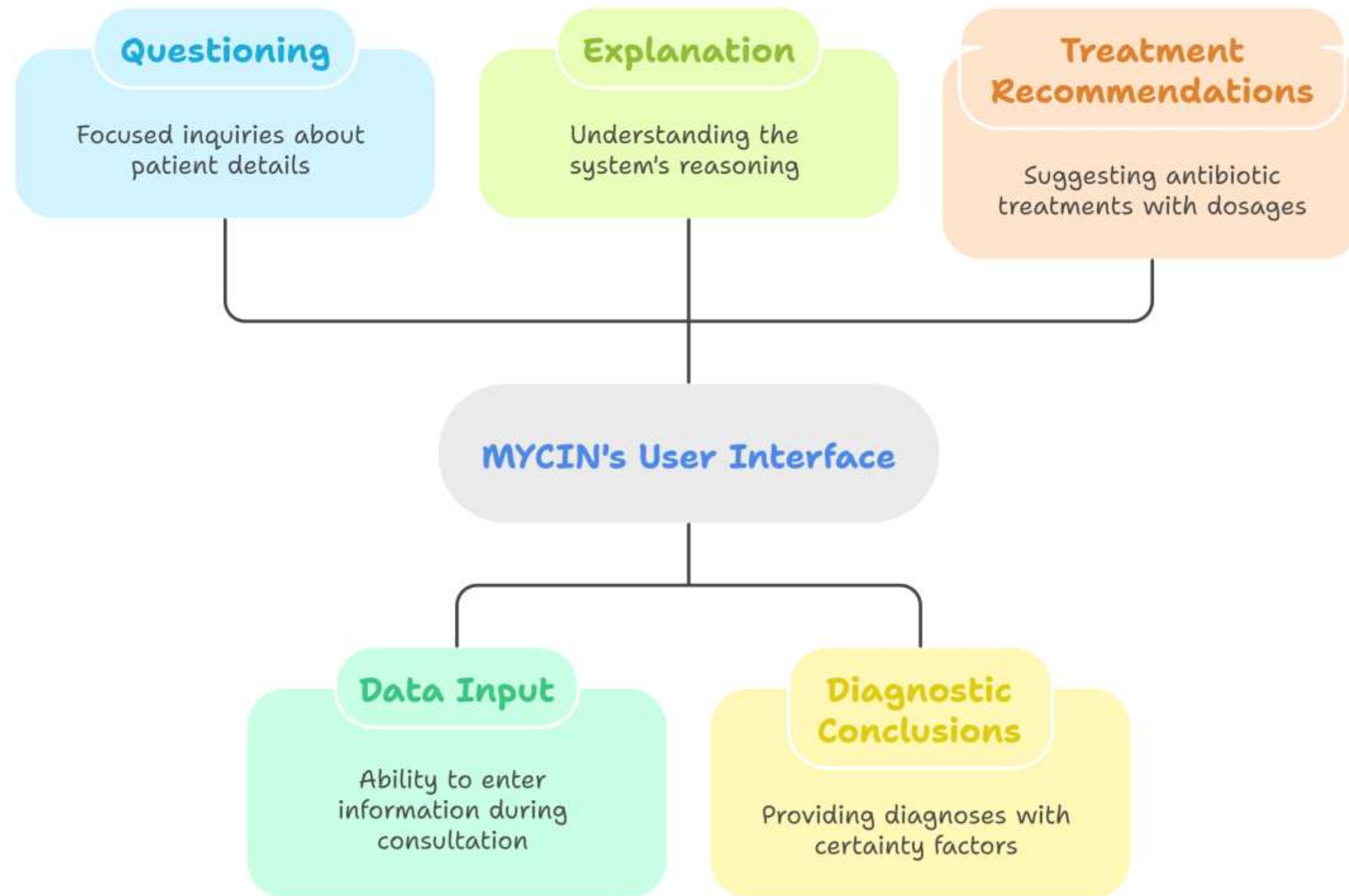
MYCIN's Inference Engine



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The engine could explain its reasoning by tracing back through the rules it applied, showing doctors why it reached particular conclusions.

MYCIN's User Interface Overview



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MYCIN: Is the patient's illness with organism-1 a hospital-acquired infection?

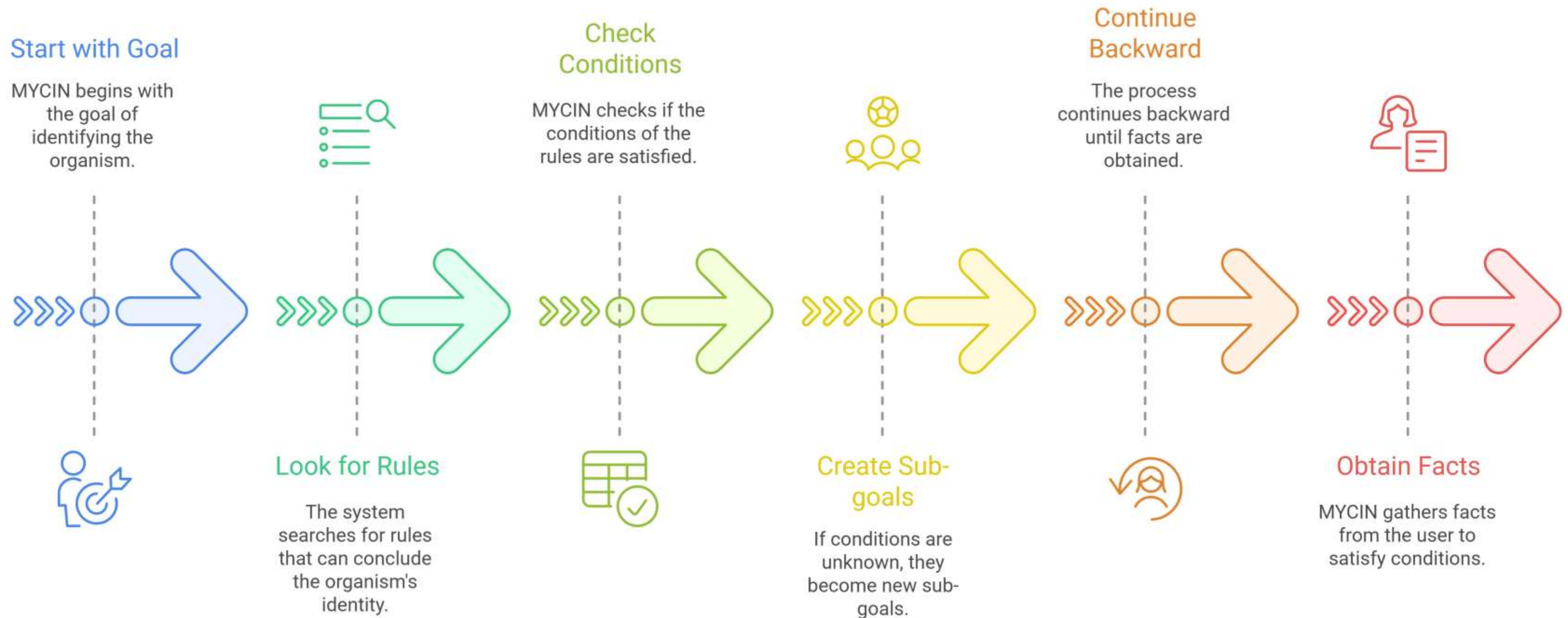
USER: Yes

MYCIN: Do you know the identity of organism-1?

USER: No

MYCIN: Is organism-1 a rod or coccus?

MYCIN's Backward Chaining Process



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GOAL: Identify the organism

Rule: IF (infection is primary-bacteremia)
AND (site is blood)
AND (patient is compromised-host)
THEN organism is likely E.coli (0.8)

Sub-goals created:

- Is infection primary-bacteremia? → Ask user or check other rules
- Is site blood? → Ask user
- Is patient compromised-host? → Ask user or check other rules

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9.1.2 Characteristics of an Expert System

High Performance:

Solves complex domain problems with high efficiency and accuracy.

Dependent on a well-updated knowledge base.

Understandable:

Results are provided in a human-understandable format. Takes input and gives output in natural language.

Reliable:

Generates efficient and accurate output consistently.

Highly Responsive:

Provides fast results, faster than human experts.

No Memory Limitations:

Can store and process huge amounts of data.

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9.1.6 Advantages of Expert Systems

- **Reproducible Results:** Results are consistent and can be reproduced each time.
- **Safe in Risky Environments:** Can be used in hazardous places where human presence is risky.
- **Accuracy:** Provide accurate results, especially with regularly updated knowledge bases.
- **Consistent and Fast Performance:** They perform consistently without being affected by emotions or fatigue.
- **Easily Available:** Mass-produced software makes expert systems widely available.
- **Affordable:** Production costs are reasonable, making them cost-effective.
- **High Speed:** Operate at high speed, reducing human effort and time.
- **Explainable Results:** Capable of explaining how conclusions or solutions were reached.
- **Improved Decision Quality:** Enhance the quality of decisions made.
- **Efficient Use of Expert Knowledge:** Efficiently gather and use scarce expertise for better solutions.

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9.1.7 Limitations of Expert Systems

Wrong Results: Incorrect or outdated knowledge leads to wrong results.

Different Solutions: Human experts may give different answers from the system.

Lack of Creativity: Can't handle unusual or creative scenarios.

High Cost: Expensive to develop and maintain. Hard to

Gather Knowledge: Collecting the right knowledge is difficult.

Depend on Experts: Success relies on expert knowledge input. **No Self-Learning:** Can't learn from experience, needs manual updates.

Needs Plenty of Knowledge: Can't work with limited information.

- **Training Intensive:** Requires extensive training to function well.
- **No Emotions:** Cannot understand or use emotions in decisions.
- **No Common Sense:** Lacks human-like reasoning.
- **Domain-Specific:** Only works for a specific area.
- **Struggles with Unusual Situations:** Can't make decisions in rare cases.
- **Garbage In, Garbage Out (GIGO):** Poor data leads to poor results. **High Maintenance Cost:** Regular updates and maintenance are expensive.

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9.1.8 Applications of Expert Systems

Design & Manufacturing: VLSI systems, camera lenses, automobiles.

Knowledge Publishing: Tax advisors and other advisory systems.

Finance: Detect fraud, suspicious activities, stock trading, loan approvals.

Medical Diagnosis: Used extensively for diagnosing diseases and treatment.

Planning & Scheduling: Project goals, airline scheduling, cargo, manufacturing.

- **Software Development:** Tracking project progress.
- **Monitoring Systems:** Detect leaks in pipelines, continuous data comparison.
- **Process Control:** Control physical processes through monitoring.
- **Automobile & Electronics:** Fault detection in vehicles and computers.
- **Complex System Diagnosis:** Electronic, electromechanical, diesel-electric systems.

Thank You Module4

- Completed